

Towards an Ontology-Based Recommender System for the Vehicle Sales Area

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Abstract. In the vehicle sales area, a recommender system (RS) allows vehicle items to be displayed and selected replied on the parameters that are the most relevant to specific users-buyers. In other words, users-buyers can meet with vehicle items suited to their profile, preferences, or interests. Therefore, users will not spend much time searching and retrieving what corresponds to their needs and capacities, as well as against the ever-growing volume of information in the application. In this paper, we realize the analysis of the characteristics of users-buyers and vehicle items in vehicle recommendations. As a result, we propose our approach for building a RS based on ontologies in the vehicle sales area.

Keywords: Recommender system · Ontology · Vehicle recommendation

1 Introduction

In today's era of instant information, more users[1] around the world are becoming more reliant upon Recommender Systems (RSs) to have better advice, suggestions, and inspire them. From e-commercial systems to online advertisements, RSs increasingly influence user's decisions to choose products or services and spend their money, time. Besides, they help users to enhance their experiences on different platforms. These ambitions have been fulfilled with the directly integrated module on their online system and they bring positive evaluations and experiences from users. These systems nudge users to explore new opportunities and suggestions matching their preferences that may never have crossed their minds [20].

In response to this trend, we desire to improve and strengthen user's interactions and experiences by a recommendation engine that will be able to assist users in searching for a vehicle as well as suggesting relevant vehicles to them according to their preferences, history, and needs. The research, analysis, and

[1] We use the term "User" instead of "User-Buyer" in the rest.

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construction of an effective RS for the purchase/sales of vehicles thus becomes our main objective.

The use of knowledge bases as ontology in the RSs has emerged in terms of the vehicle sales area [16,18]. However, these models have focused on finding and matching users' interest and vehicle items descriptions based on keywords or terms, and semantic relations between these keywords or terms. In this paper, we investigate to go beyond applying ontologies not only to finding and matching data but also to reasoning information and computing similarity between user profiles based on ontologies.

The remainder of this paper is organized as follows. Section 2 studies related works. Section 3 presents our contributions in (i) proposing architecture for the development of an ontology-based recommender system in the vehicle sales area, (ii) inheriting and developing user profile and vehicle ontologies, (iii) calculating similarities between user profile and vehicle items. We present a use case example to illustrate our approach in Sect. 4. Finally, we conclude the paper with remarks for future work in Sect. 5.

2 Related Work

In this section, we introduce the fundamental knowledge about ontologies and recommender systems. In addition, we summarize related work concerning ontologies, RSs, and the use of the ontologies in RSs.

In the context of information science and knowledge sharing, ontology is a form of knowledge representation that consists of a set of concepts within a domain and the relationships between those concepts [6]. Ontology is based on an explicit formal description that allows to shared common understanding of the structure of information, to reuse of domain knowledge, to separate domain knowledge from operational knowledge. The usefulness of ontologies is relied on some reasons as organizing data, improving search, and data integration [2].

One of the purposes of ontology is to capture knowledge about a domain and represent them in a machine-readable and interpretable form. Various ontologies are designed for knowledge representation in the vehicle sales area. More specially, ontologies can be developed to improve semantic search [5]. Or ontologies provided the vocabularies for describing the vehicle such as the Vehicle Sales Ontology (VSO)[2] and the Used Cars Ontology (UCO)[3]. In general, these ontologies are applied to increase the visibility of a vehicle in search engines. On the other sides, the user profile ontology has been widely designed and used in many different applications. The majority of user profile ontologies include set terminology that can describe all characteristics of a user [3].

RS conventionally is an application that attempts to recommend the most relevant items to users by reasoning or predicting the user preferences in an item replied on related information about the users, the items and the interactions between items and users [14]. In general, RSs can be classified into demographic-based RSs, Content-based RSs, Collaboration Filtering-Based (CF-Based) RSs, Knowledge-based RSs, Context-aware RSs, and Hybrid RSs.

[2] http://www.heppnetz.de/ontologies/vso/ns.
[3] http://ontologies.makolab.com/uco/ns.html.

The construction of recommender systems based on using ontologies as a knowledge base has been implemented and experimented within various domains such as course recommendations in e-learning systems [8,17], tour recommendations in tourism applications [11], or product recommendation in e-commercial systems [7]. The use of ontology in the RSs has been developed in the vehicle sales area. For example, the work by [18] has proposed an ontology-based RS that uses an ontology-based hierarchical layered architecture in order to improve semantic searching with understanding hierarchical relations of concepts. *Mu et al.* [16] applied to match between vehicle ontologies and user's interest in order to give the most suitable vehicles corresponding with users' interest.

3 Ontology-Based Vehicle Recommender System

3.1 Needs for Building an Explainable RS in the Vehicle Sales Area

Knowledge bases can be employed to construct an explainable intelligent system. Based on the fundamentals of mathematical logic, systems using a knowledge base can make deductive reasoning provide new information through mathematically proven ways. The construction of an explainable RS in the vehicle domain needs also be carefully considered in the case of helping users better understand the results they receive.

The lack of data directly affects choosing a RS approach as well as relates to the way the evaluation and experiments of the RS. Therefore, gathering and preprocessing data is an indispensable step in the construction of any RS. More specifically, we need to determine which types of data are important and the way how these data will be collected. Besides the organization and management of data are also important factors to be scrutinized.

With RSs, each approach always exists advantages and inconveniences. Using only one approach will benefit about time and cost for the construction of a RS. However, the developed RS will exist shortcomings in some aspects. Therefore, a hybrid system can be a reasonable choice to trade off the limitations of each approach. Although deep neural network models are playing increasingly crucial roles across a wide range of decision-making problems [10]. However, the use of the content-based or the CF-based approaches associated with the deep learning approach perhaps is not the most suitable option in the current condition of data. With special characteristics in the vehicle sales area, the use of knowledge base as ontology in the combination with the content-based and the CF-based approaches in a hybridization design reaches an acceptable solution.

3.2 Development of an Ontology-Based Vehicle Recommender System

The proposed approach is a hybrid recommender system that is combined collaborative filtering, content-based approaches, and ontologies. A combination of both content-based and CF approaches limits shortcomings about cold start or data sparsity problems. Besides using ontology reinforces the accuracy in computing semantic similarity measures about vehicle descriptions, user profiles,

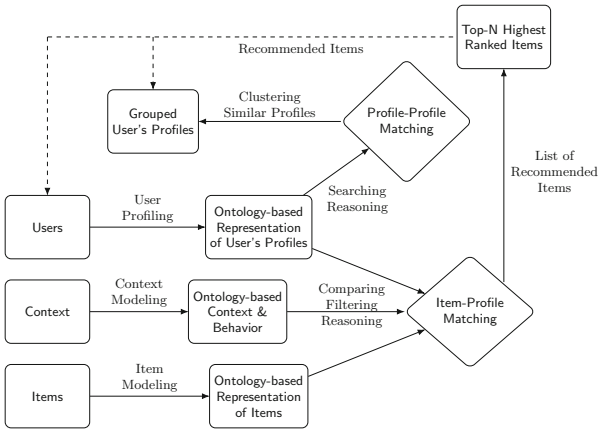


Fig. 1. The recommendation process of the proposed recommender system

and contextual information. As shown in Fig. 1 the process is divided into five steps started by gathering and modeling data to find the recommendation items for each special user, and terminated by filtering and ranking recommendation results as follows:

- Gathering data from different sources and building a knowledge base for users, vehicle items, context, and rating data of items based on ontologies.
- Organizing and mapping different ontologies to link the user profiles, context, and item profile descriptions based on the existing data, mapping the relational database to RDF triples for user profile, item descriptions, contextual information, and interaction data of users for items.
- Filtering and reasoning vehicle items that matched user preferences to reduce computation space.
- Using the content-based approach, calculating the similarity between user preferences and vehicle items by feature vectors of user preferences and vehicle items.
- Using the collaborative filtering approach, calculating the similarity of user recommendation history and similarity between the user profile by using ontology-based semantic similarity.
- Ranking and filtering the list of relevant vehicle items to a user.

3.3 Data Gathering

The limitation of the amount of user and vehicle data is always a challenge in the domain of constructing a recommender system. In our case, the extraction of information will come from both the user and the vehicle. We explore users' information from their profile which is collected from questionnaires and surveys. Besides demographic information, user preferences play also an important role in deeper understanding the intention of users in the decision of purchasing a

vehicle. In that regard, we focus user preferences on vehicle color, brand, mileage, budget, the number of seats, and so on.

The vehicle description includes the permanent registration published on open databases such as the Vehicle Identification Number (VIN) which provides manufacturer, portrait of the vehicle, engine size, and type, fuel type, the model year of the vehicle. In addition, the current situation and condition of the vehicle such as the date and expiry of mandatory optional certifications and inspections, non-standard vehicle features, damage or accident information, and others should be collected to have a general view about the vehicle.

3.4 Ontology Construction

This section focuses on constructing knowledge bases with ontology and data sources under triplet sets. The present work allows obtaining two ontologies about user profile and vehicle ontology. Firstly, the user profile ontology model provides a general view of a user and their preferences. We then organize and structure vehicle descriptions with a vehicle ontology.

User Profile Ontology. Many works have constructed user profile ontologies for the applications in various domains from education [1,8], e-commerce [13], to tourism [15]. We develop the User Profile Ontology based on User Profile Ontology (UP)[4]. Besides demographic and social information about a user such as gender, year old, address, occupation, income's range, language, number of kids, married, education level (see Fig. 2), our user profile ontology takes into

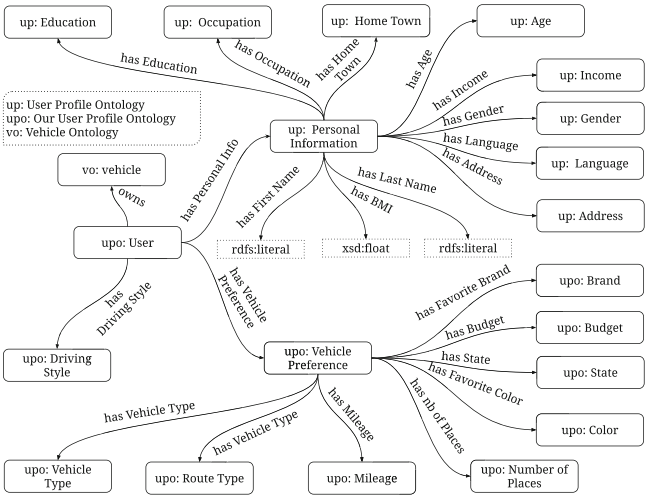


Fig. 2. A visualization of the user profile ontology

[4] http://iot.ee.surrey.ac.uk/citypulse/ontologies/up/up.html.

account psychological information about user preferences that directly influence the consideration and choice of a user to purchase a vehicle. In general, we consider important preferences such as budget, vehicle brand, number of seats, mileage, vehicle type, favorite color, vehicle state, route type.

By using this ontology, the profile of users is organized under triplets in a graph representation. Therefore, a user profile can be formally defined as follows:

$$G_U = \{a_1^u, a_2^u, ..., a_n^u\} \tag{1}$$

where a_i^u denotes the triplet $a_i^u = <subject_i, predicate_i, object_i>$. In other words, the triplet a_i^u can also be expressed as $<resource_i, propertie_i, statement_i>$. For example, "*Henri is interested in blue cars*". This information can be represented in a triplet as $<Henri, hasFavoriteColor, Blue>$. Therefore, a set of triplets of describing a user creates a graph representation where the nodes is equivalent to resources or subject, and arcs denotes named properties.

Vehicle Ontology. Different ontologies have been developed in order to describe and model vehicle information and apply them in several applications. For instance, automobile ontologies have been designed to manage semantic knowledge between the user, the vehicle, and the current driving situation [4,9]. Or an ontology has been proposed to describe tracking, monitoring, and diagnosing problems of an automobile [19]. In our case, the vehicle ontology is based on two ontologies: VSO and UCO. These ontologies provide a standard vocabulary for describing aspects of new and used vehicles for e-commercial applications. Therefore, the vehicle ontology allows a combination of modeling the permanent information such as vehicle identify, license plate number, manufacturer, model, and so on. Besides we can describe status changes of a vehicle such as damage, accident, inspections, mileage (see Fig. 3).

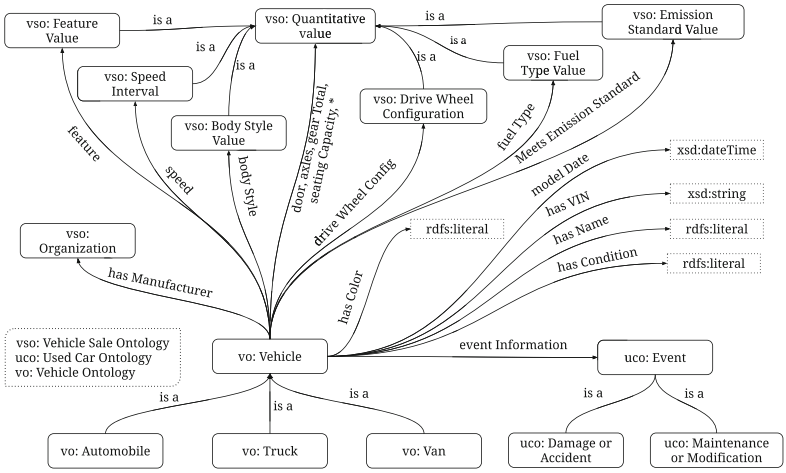


Fig. 3. A visualization of the vehicle ontology

A vehicle description can be represented as a knowledge graph. Similarly, with the user profile, we can define a vehicle description model as follows:

$$G_V = \{a_1^v, a_2^v, ..., a_k^v\} \tag{2}$$

where a_i^v presents the triplet $a_i^v = <subject_i, predicate_i, object_i>$ or $a_i^v = <resource_i, propertie_i, statement_i>$.

When a user makes an interaction with the vehicle by giving a rating, commenting, or adding to a favorite list, we mark these interactions to have an analysis of the intention and behavior of the user and offer relevant recommendations. Therefore, we can further define these interactions as follows:

$$G_I = \{G_U, G_V, r\} \tag{3}$$

where r is the interactive way of the user G_U on the vehicle G_V.

3.5 Semantic Recommendations from Filtering and Reasoning

Ontologies play a key role in the organization and exploitation of knowledge. Ontology provides a common vocabulary and grammar for data as well as semantic relation and description of the data. To develop a knowledge base replied on principles of ontology, different techniques are employed in a combination way from organizing data (e.g., RDF, RDFS, OWL), to querying data (e.g., SPARQL), and to inferring explicit knowledge (e.g., SWRL).

We started with a definition of SWRL rules to inference or calculate information from user profiles. For example, as in the Listing 1.1, we can infer the age of a user from their birthday with two steps: the rule R1 calculates the year born by using the SWRL function *swrl.date*, then the rule R2 uses the *swrl.subtract* function to the year born and the current year to get the age of the user. The rule R3 allows to infer whether a user is a driver or non-driver by comparing the age with the *swrl.greaterThanOrEqual* function.

```
R1 - User(?x), hasBirthday(?x, ?date), swrlb.date(?date,
 ?year, ?month, ?day, ?timezone) => bornInYear(?x, ?year)
R2 - User(?x), bornInyear(?x, ?year), swrlb.subtract(?age,
?nowyear, ?year), thisYear(?nowyear)  => hasAge(?x, ?age)
R3 - User(?x), hasAge(?x,?age), swrlb.greaterThanOrEqual(?age,
18), owns(?x,?vehicle) => isDriver(?x,True)
```

Listing 1.1. Age and driver inference by using the SWRL Rules

In order to make a relevant recommendation, it is important to ensure that the characteristics of vehicle items match the preferences of the user. User profiles and Vehicle descriptions are stored in a triplestore that allows to implementation of efficient querying of a graph-based database. Therefore, we define SPARQL queries to filter and retrieve a list of vehicle descriptions that has the most common characteristics with the user's preference declaration. For example, as in the Listing 1.2, we declare a SPARQL query to retrieve user profiles and vehicle descriptions that have the same favorite color and color of the vehicle.

```
PREFIX vo: <http://vivocaz.fr/vo/ns#>
PREFIX upo: <http://vivocaz.fr/upo/ns#>
SELECT ?user ?vehicle
WHERE {
?user upo:hasPreference ?userPreference .
?userPreference upo:hasFavoriteColor ?color .
?vehicle vo:hasColor ?color .
} ORDER BY ?user LIMIT 10
```

Listing 1.2. A SPARQL query using to search vehicles by the user's favorite color

3.6 Recommendation Computation

In this section, we investigate applying content-based and collaboration filtering approaches to calculating similarities that exist between vehicle items and user preferences as well as between user profiles.

User preference and vehicle item similarity is started with transforming user's interest and vehicle description features to vectors. The user preferences include features as *route type*, *state*, *mileage*, *vehicle type*, *color*, *profile type*, *brand*, *budget* and *number of seats*. These features need to match with vehicle description. A similarity measure is determined to describe the proximity of two vectors. In the simplest case, a cosine similarity can be employed as follows:

$$Sim(G_{U_i}, G_{V_j}) = \frac{\vec{G}_{U_i}.\vec{G}_{V_j}}{\|\vec{G}_{U_i}\|.\|\vec{G}_{V_j}\|} \tag{4}$$

where G_{U_i}, G_{V_j} denote user preferences and vehicle descriptions. Hierarchy ontology similarity can be used to add feature vectors because it helps to add more information based on means of conceptual semantic and lexical relations such as hypernymy, meronymy, and synonym.

User profile similarity allows finding a list of users that have the most similar characteristics with the users who have various interactive activities such as rating, having favorite vehicle item list on the applications. The similarity between the target user and other users is measured by ontology-based semantic similarity $Sim_O(G_{U_i}, G_{U_t})$ based on user profiles and item recommendation history similarity $Sim_H(G_{U_i}, G_{U_t})$ as follows:

$$Sim(G_{U_i}, G_{U_t}) = Sim_O(G_{U_i}, G_{U_t}) + Sim_H(G_{U_i}, G_{U_t}) \tag{5}$$

where G_{U_i} and G_{U_t} denote the user profile i and the target user profile t.

Dive into ontology-based semantic similarity, it can be calculated from feature-based semantic similarities correspondence with using *Object* in a triplet or from information content-based semantic similarities that is equivalent to

using *Predicate*. In our case, we use the formula proposed in [12] in order to calculate this index that is defined as follows:

$$Sim_O(G_{U_i}, G_{U_t}) = \sum_{g=1}^{P^{G_{U_i}, G_{U_t}}} \frac{|O_{p_g}^{G_{U_i}} \cap O_{p_g}^{G_{U_t}}| \times \frac{IC(P^{G_{U_i}})}{\sum_{k=1}^{K} IC(P^{G_{U_i}})}}{|O_{p_g}^{G_{U_i}} \cap O_{p_g}^{G_{U_t}}| + \alpha|O_{p_g}^{G_{U_i}} - O_{p_g}^{G_{U_t}}| + \beta|O_{p_g}^{G_{U_t}} - O_{p_g}^{G_{U_i}}|} \quad (6)$$

where $P^{G_{U_i}, G_{U_t}}$ denotes the number of predicates existing in both G_{U_i} and G_{U_t}, p_g is a predicate of G_{U_i} and G_{U_t}, $O_{p_g}^{G_{U_i}}$ indicates a set of qualitative objects related through the predicate p_g, $|O_{p_g}^{G_{U_i}} \cup O_{p_g}^{G_{U_t}}|$ denotes the number of common objects between G_{U_i} and G_{U_t} through the predicate p_g, $|O_{p_g}^{G_{U_i}} - O_{p_g}^{G_{U_t}}|$ represents the number of different objects of $O_{p_g}^{G_{U_t}}$ in $O_{p_g}^{G_{U_i}}$, $IC(P^{G_{U_i}})$ is the amount of information provided by the predicates that G_{U_i} contains, $\sum_{k=1}^{K} IC(P^{G_{U_i}})$ is the sum amount of information provided by the predicates in G_{U_i}, and $\alpha, \beta(\geq 0)$ expresses the weights of G_{U_i} and G_{U_t}.

With the calculation of user's recommendation history similarity, we use also cosine similarity algorithm which is defined as follows:

$$Sim_H(G_{U_i}, G_{U_t}) = \frac{\sum_{k \in R} G_{U_i}.G_{U_t}}{\sqrt{\sum_{k \in R} G_{U_i}^2} \cdot \sqrt{\sum_{k \in R} G_{U_t}^2}} \quad (7)$$

where R is the set of vehicle items that have been interacted by user G_{U_i}, G_{U_t}.

4 Use Case Example

We illustrate a use case example in order to show how our approach works. From the ontologies, we collect maximal information about the user and vehicles and represent them in triplets. For example, we have the triplets of the user Henri <*Henri, hasPersonalInformation, HenriPI*>, <*HenriPI, hasOccupation, Doctor*>, <*HenriPI, hasBirthday, 04/19/1975*>, <*Henri, hasVehiclePreference, HenriVP*>, <*HenriVP, hasFavoriteColor, Blue*>, <*HenriVP, hasVehicleType, Sedan*>, and <*Henri, owns, TeslaModelS01*>.

By using the rules SWRL, we can infer information around a user. For example, we have information about the age of Henri and identify that the user is a driver or a non-driver based on the rules defined in Listing 1.1. Furthermore, we can filter or query users and vehicles which have common characteristics with Henri by query language SPARQL. More especially, we search vehicles which have the characteristics matched with Henri preferences such as <*RenaultClio01, hasName, Renault Clio 2021*>, <*RenaultClio01, hasColor, Blue*>, <*KiaRio01, hasName, Kia Rio 2021*>, <*KiaRio01, hasColor, Blue*>, <*VolkswagenJettaGLI01, hasName, Volkswagen Jetta GLI 2021*>, <*Volkswagen JettaGLI01, hasColor, Blue*>.

The recommendation vehicle items list is calculated based on the similarity between the user preference and vehicle items. First, the set of triplets achieved

from SPARQL queries is used to extract a set of features. These features are then transformed into vectors by using some techniques as Tf-Idf or Bag-of-Words. Finally, the similarity between the user's vehicle preferences and vehicle items is calculated by using the formula 4.

We can add into the recommendation vehicle items list from the calculation of the similarity between user profiles and user's recommendation history by employing the formula 5, 6, 7. For example, a user Louis have information as *<Louis, hasPersonalInformation, LouisPI>*, *<LouisPI, hasOccupation, Doctor>*, *<LouisPI, hasBirthday, 01/12/1985>*, *<Louis, hasVehiclePreference, LouisVP>*, *<LouisVP, hasFavoriteColor, White>*, *<LouisVP, hasVehicleType, Sedan>*, and *<Louis, owns, Peugeot20801>*. Using ontology-based semantic similarity defined in formula 6, we can obtain the similarity between the user Henri and Louis. In addition, we calculate the user's recommendation history with the formula 7. Finally, the ranking score of items gives an order of recommendation items for the user.

5 Conclusion and Perspective

The objective of this study is the development of a RS that can generate personalized vehicle recommendations matched with the needs and capacities of users. Using ontologies of user profiles and vehicle descriptions allows enriching information as well as semantic relationships between users, between vehicles, and between users and vehicles. We compute the similarities by using the score functions in order to rank obtained results. Ontology-based semantic similarity calculation is part important in our approach because we can apply it to our ontology model in various aspects both users and vehicles. Furthermore, we can consider a towards explainable recommendation results in which users can understand why a vehicle item is the right suggestion in their case.

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